

# No Man's Land

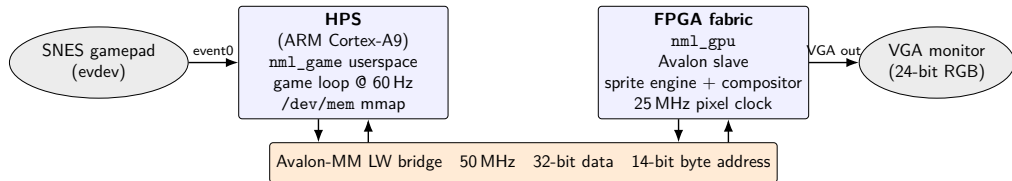
A custom SystemVerilog GPU and game on the DE1-SoC

Rohit Biswas   Kambinachi Obioha   Nicola Paparella

CSEE 4840 — Embedded System Design — Spring 2026

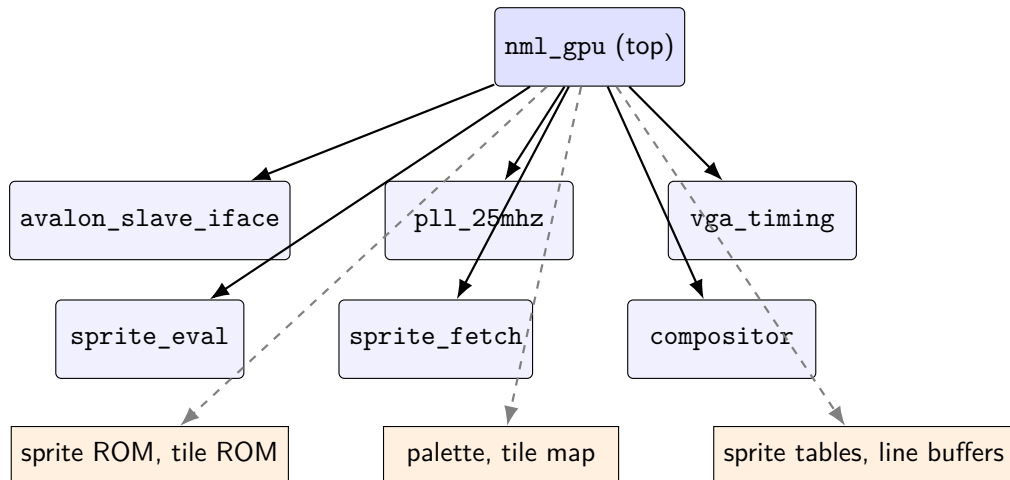
Instructor: Prof. Stephen Edwards

# System block



HPS writes shadow state; FPGA latches at VSYNC; pixels always stream.

# What's inside nml\_gpu

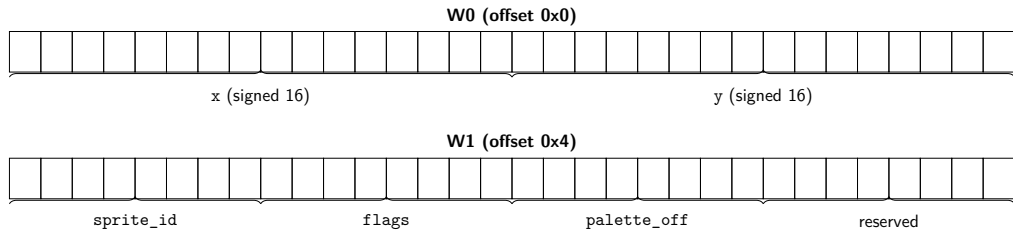


One 50 MHz domain (Avalon) + one 25 MHz domain (pixel). All RAMs/ROMs inferred as M10K.

# Avalon register map — 16 KB window at 0xFF200000

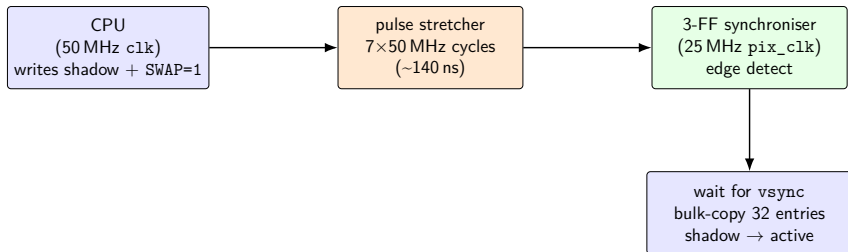
| Offset        | Name                | Layout                                        |
|---------------|---------------------|-----------------------------------------------|
| 0x0000        | CTRL                | [0]=ENABLE, [1]=SWAP (auto-clear), [2]=HUD_ON |
| 0x0004        | STATUS              | [0]=VBLANK, [1]=SWAP_PENDING                  |
| 0x0008        | BG_SCROLL           | [15:0]=scroll_x, [31:16]=scroll_y (signed)    |
| 0x000C        | IRQ_MASK            | [0]=VBLANK IRQ enable                         |
| 0x0010        | PLAYER_POS          | [15:0]=px, [31:16]=py                         |
| 0x0014        | PLAYER_STATS        | [7:0]=hp, [15:8]=wave (BCD), [31:16]=level    |
| 0x0018        | SCORE               | [23:0]=6-digit BCD score                      |
| 0x001C        | KILL_COUNT          | 32-bit kills                                  |
| 0x0020        | HUD_AUX             | [7:0]=ammo BCD, [11:8]=art, [15:12]=gas       |
| 0x0100-0x01FF | <b>Sprite table</b> | 32 entries × 8 B                              |
| 0x0400-0x07FF | <b>Palette</b>      | 256 entries × 4 B (RGB888)                    |
| 0x1000-0x22BF | <b>Tile map</b>     | 80 cols × 60 rows, 1 byte/tile                |

## Sprite-table entry — 8 bytes / 64 bits



**flags byte:** [2] hflip [3] vflip [6:4] priority (0 = highest) [7] ACTIVE

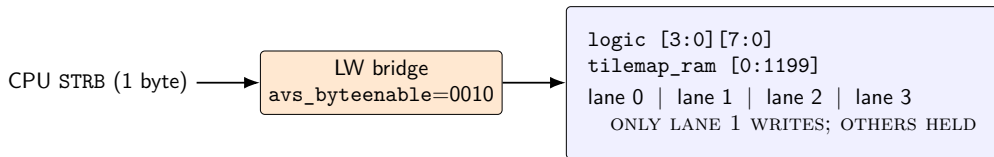
# Frame commit — shadow → active at VSYNC



CPU then polls `STATUS.SWAP_PENDING`;  
clears one cycle after the bulk copy.

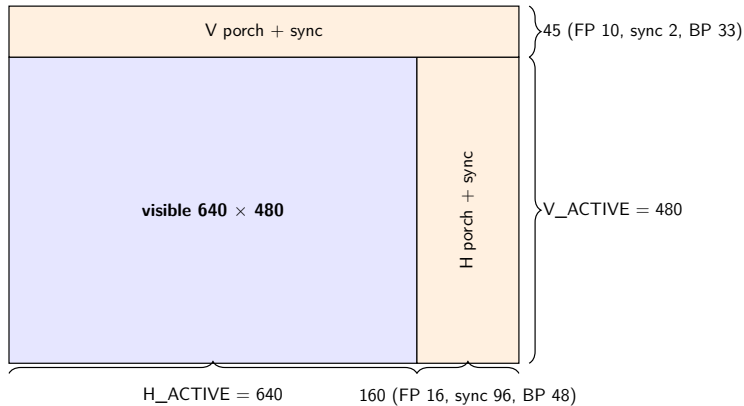
One 20 ns Avalon pulse would slip past the 40 ns pixel clock. Widen to 140 ns and the CDC is deterministic.

# Tile-map byte writes — packed [3:0] [7:0]



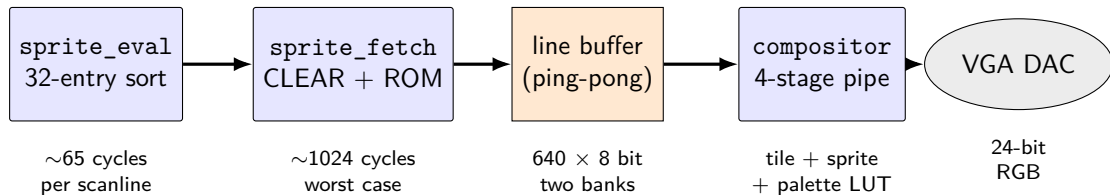
**Why packed?** Byte-organised storage forced bridge RMW to clobber 3 lanes per word. Packed inference uses M10K `byteena_a`: one lane updates, the other three are untouched.

## VGA timing — $640 \times 480$ @ 60 Hz, 25 MHz dot



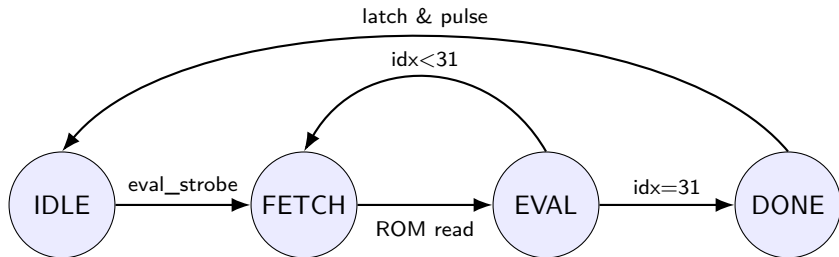
One frame =  $800 \times 525$  dot cycles. HBLANK is the 160-cycle window we use to evaluate sprites for the next scanline.

## Pixel pipeline — sprite engine + compositor



Eval runs inside HBLANK (160 cycles). Fetch spans HBLANK + visible (800 cycles). Line buffers are double-buffered, so fetch lags read by one scanline. Compositor never stalls.

## sprite\_eval — per-scanline priority sort



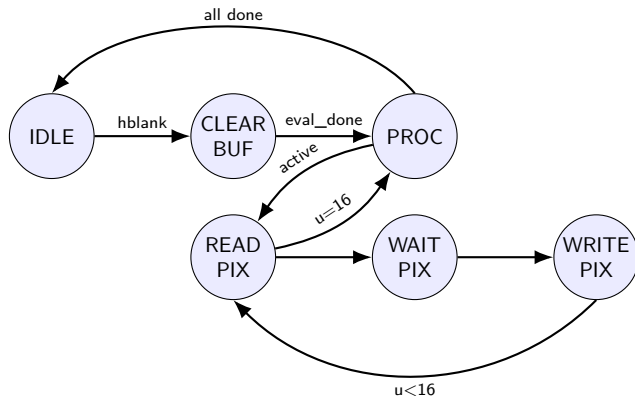
Per sprite (1 cycle in EVAL):

`intersects = active && y ≤ next_scanline < y+16`

insertion-sort into 8 priority-ordered slots (all 8 compares in parallel)

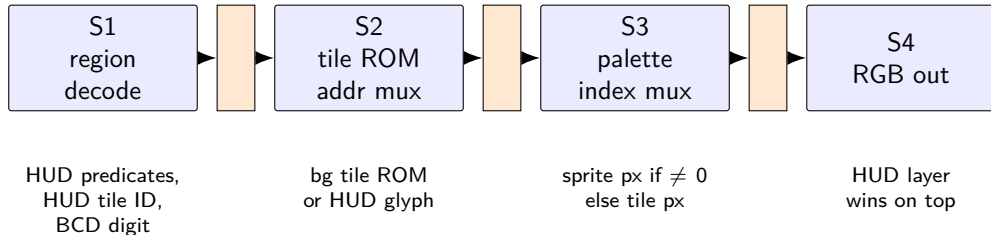
Total: 32 fetch+eval cycles + 1 DONE = ~65 cycles. Fits easily in 160-cycle HBLANK.

## sprite\_fetch — ROM read + line-buffer fill



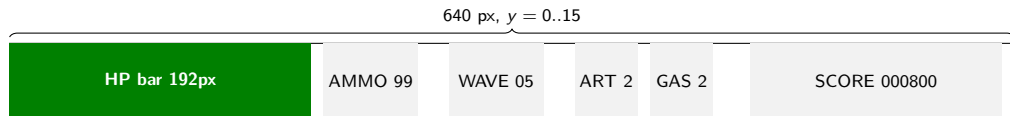
**Worst-case cycle budget per scanline:**  $\text{CLEAR\_BUF (640)} + 8 \text{ sprites} \times 48 \text{ (384)} \approx 1024$  cycles.  
HBLANK alone is 160; HBLANK + visible is **800**. Fetch lives one line behind the read, so the ping-pong line buffers absorb the latency.

# Compositor — 4-stage pipeline



Latches between stages absorb the 1-cycle latency of tile ROM and palette M10K reads. Never stalls.

# HUD strip — top 16 pixels of every frame



HP bar: 2 px per HP, clamped to 192. Glyphs: 8×8 from tile ROM, slots 16–41 (A–Z), 48–57 (0–9). Scores, ammo, wave packed BCD in registers.

## Memory inventory (all inferred M10K)

| Memory            | Shape                                    | Init                       | Bits               |
|-------------------|------------------------------------------|----------------------------|--------------------|
| sprite ROM        | $16,384 \times 8$                        | \$readmemh                 | 131,072            |
| tile ROM          | $4,096 \times 8$                         | \$readmemh                 | 32,768             |
| tile map          | $1,200 \times 32$ (packed $4 \times 8$ ) | blank (HPS writes)         | 38,400             |
| palette           | $256 \times 24$                          | \$readmemh + HPS overrides | 6,144              |
| sprite table act. | $32 \times 64$                           | via VSYNC bulk copy        | 2,048              |
| sprite table shd. | $32 \times 64$                           | HPS writes                 | 2,048              |
| line buffer A     | $640 \times 8$                           | cleared per HBLANK         | 5,120              |
| line buffer B     | $640 \times 8$                           | cleared per HBLANK         | 5,120              |
| <b>Total</b>      |                                          |                            | <b>86,144 bits</b> |

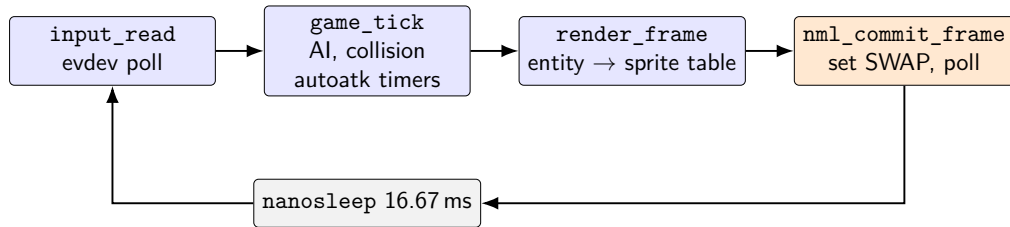
14 M10K blocks out of 397.

## Resource utilisation — Cyclone V 5CSEMA5F31C6

| Resource          | Used   | Available | %    |
|-------------------|--------|-----------|------|
| ALMs              | 205    | 32,070    | < 1% |
| Registers         | 365    | —         | —    |
| Pins              | 54     | 457       | 12%  |
| Block memory bits | 86,144 | 4,065,280 | 2%   |
| M10K blocks       | 14     | 397       | 4%   |
| DSP blocks        | 0      | 87        | 0%   |

Compile time: 63 s wall clock (Analysis 13 s, Fitter 30 s, Asm 12 s, STA 8 s).

# Software architecture — 60 Hz game loop on the ARM



game\_t holds a 64-entity pool. State machine: PLAYING ↔ LEVELUP → GAMEOVER.  
Mortar is timer-driven; artillery + gas fire from L/R shoulders.

# What actually works

---

640×480 @ 60 Hz VGA on the DE1-SoC monitor  
Tile-mapped battlefield (80×60 from 64-tile ROM)  
32 hardware sprites, top-8 sorted per scanline, hflip  
256-entry RGB888 palette, M10K-inferred  
HUD strip: HP bar, AMMO, WAVE, ART, GAS, SCORE (BCD)  
DragonRise SNES gamepad over evdev (/dev/input/event0)  
60 Hz HPS game loop driving sprite-table writes  
Frame-commit handshake (CTRL.SWAP → VSYNC bulk copy)  
Mortar timer, artillery column kill + beam, mustard gas cloud  
Wave spawner, level-up menu, ammo drops, game-over screen

---